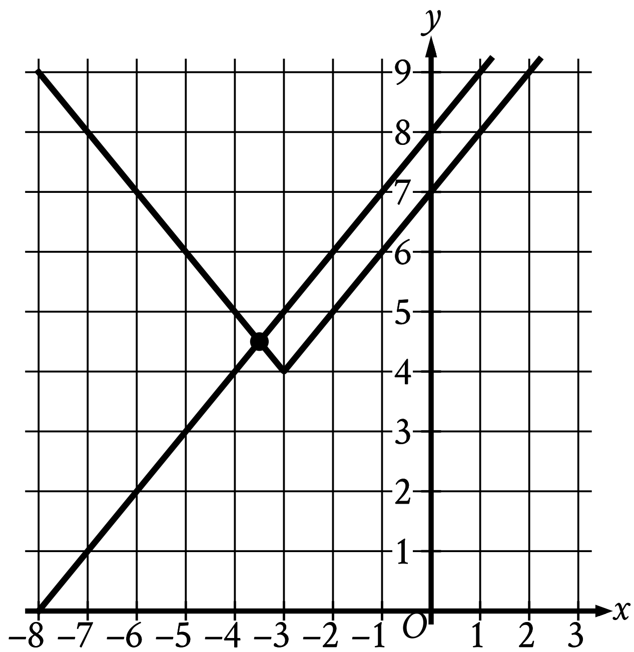
Question ID: 494d247d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Easy

Question



The graph of a system of an absolute value function and a linear function is shown. What is the solution (x, y) to this system of two equations?

Answer

A. $(0, 8)$

B. $\left(\frac{7}{2}, \frac{9}{2}\right)$

C. $\left(-\frac{7}{2}, \frac{9}{2}\right)$

D. $(-3, 4)$

Correct Answer: C

Rationale

Choice C is correct. The solution to a system of two equations corresponds to the point where the graphs of the equations intersect. The graphs of the linear function and the absolute value function shown intersect at a point with an *x*-coordinate between -4 and -3 and a *y*-coordinate between 4 and 5. Of the given choices, only $\left(-\frac{7}{2}, \frac{9}{2}\right)$ has an *x*-coordinate between -4 and -3 and a *y*-coordinate between 4 and 5.

Choice A is incorrect. This is the *y*-intercept of the graph of the linear function.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the vertex of the graph of the absolute value function.